

Michelson Interferometry: Experiment I

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Physics 5700

24 Jan 2025

I. INTRODUCTION

The Michelson interferometer is an optical instrument often used to demonstrate the wave nature of electromagnetic light through the generation of interference patterns created by coherent interactions of two different wave sources. The primary function of the interferometer involves splitting a beam into two different paths of different lengths, allowing for the separate manipulation of these beams to perform various interferometry tasks (See Ref. [1]).

The Michelson interferometer was originally used in the famous Michelson-Morley experiment, which disproved the theory that light waves propagate through an invisible "ether" (See Ref. [2]). The experiment aimed to detect the relative motion of the Earth through the ether which would result in a change of the speed of light. The paper concluded that there was no detectable change in the speed of light from Earth's relative motion, disproving the existence of the ether. It also paved the way for Einstein's theory of special relativity.

The instrument is designed to provide extremely fine control over the path of each light beam. This control allows for intentional interference patterns to be observed from the recombination of the split beams. These interference patterns can be used to accurately predict beam wavelengths, wavelength separations of beam components, or the index of refraction of some transparent substance through which a beam passes.

There are three sub-experiments with the following objectives:

- 1) Measure wavelength, λ , of the Helium-Neon laser and of the Thallium green line.
- 2) Measure wavelength separation, $\Delta\lambda$, of the Mercury yellow doublet.
- 3) Measure the index of refraction, n , of a transparent material such as mica, quartz, or glass.

There are also a few secondary objectives of the experiment as a whole:

- 1) Demonstrate the wave nature of light with diffraction patterns.
- 2) Use Snell's law to manipulate the path of a beam of light.

II. SAFETY CONSIDERATIONS

This experiment requires the use of lasers, meaning that proper care must be taken to avoid harm to participants. While the lasers that are part of this lab are not powerful enough to induce blindness, it is still necessary to take precautions to limit eye and skin contact with radiation emitted by the laser.

Ohio State provides students and instructors with a required course on Laser Safety through *BuckeyeLearn*. This one hour

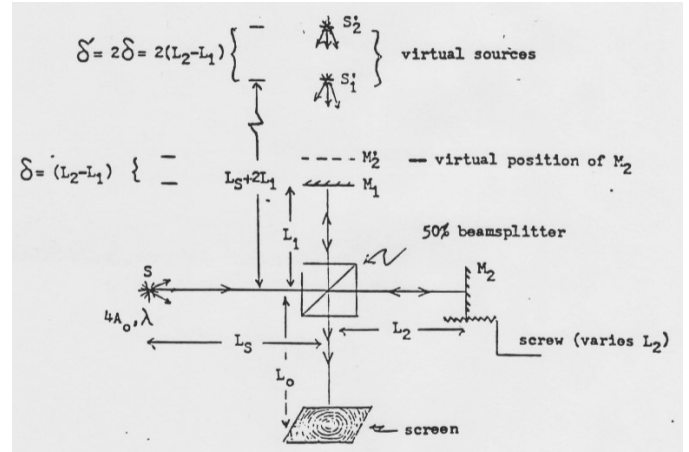


Fig. 1. Michelson Interferometer. L_S , L_O , L_1 , and L_2 are the optical paths indicated. (See Ref. [4]).

and forty minute course should be completed by all participants prior to usage of lasers in the lab (See Ref. [3]).

For this experiment, the laser beam should remain within the plane of the interferometer itself. To avoid looking into the laser, participants should avoid placing their head near the plane of the interferometer at all times, and instead observe the interferometer from above. The interferometer should be placed on a flat, stable surface. Rings, watches, and other jewelry should be removed to avoid specular reflections of the laser out of the plane of the instrument.

Standards of procedure should be followed carefully during the initial setup and alignment of the interferometer before the emitting device is fixed in place. Avoid aiming the laser upwards from the optics bench.

III. HELIUM-NEON AND THALLIUM

A. Theory

From a singular point source S , the Michelson interferometer generates two "virtual" point sources S'_1 and S'_2 via coordinated reflection and transmission of two legs of a split beam. Because of the varying path lengths of these beams, the virtual sources have a slightly different distance from the observer. The Michelson interferometer setup is shown in Figure 1. The separation between the virtual sources is given by

$$\delta' = 2\delta = 2(L_2 - L_1) \quad (1)$$

where L_1 is the distance between beam splitter and the mirror (M_1) which receives the light reflected by the beam splitter.

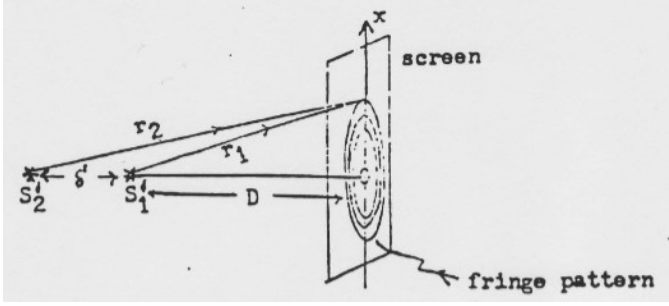


Fig. 2. Fringe pattern generated by virtual sources S'_1 and S'_2 . $D = L_O + L_S + 2L_1$; $\delta' = 2(L_2 - L_1)$. (See Ref. [4]).

L_2 is the distance between the beam splitter and the mirror (M_2) which receives the light *transmitted* by the beam splitter. Generally, there is fine motor control over one mirrors of the interferometer which allows for very slight adjustments to either L_1 or L_2 (in this case, L_2) [4].

Since the beam which follows L_2 must pass through the beam splitter three times, while the beam which follows L_1 must pass through only once, a second plate of identical thickness and material called a “compensator” is placed along the path of L_1 to eliminate the need to account for the thickness of the beam splitter [1].

The location of bright and dark fringes are given by the following equations as they relate to Figure 2.

$$|r_2 - r_1| = n\lambda \quad (2)$$

where $n = 0, 1, 2, \dots$ for bright fringes. For the dark fringes,

$$|r_2 - r_1| = n\lambda/2 \quad (3)$$

where $n = 1, 3, 5, \dots$. Since r_1 and r_2 can be calculated from known path lengths, accurate measurement of the location of the bright and dark fringes allows for an accurate prediction of the wavelength of light through these relationships (See Ref. [4]).

As the movable mirror is adjusted back and forth to change δ' , a number of rings will be seen to be either entering or exiting the center point of the “bullseye” fringe pattern. A more direct measurement of the wavelength of the source laser is given by

$$\lambda = \frac{2\Delta d}{N} \quad (4)$$

where λ is the laser wavelength, N is the number of fringes which pass by a fixed point on the screen, and Δd is the distance moved by the adjustable mirror (See Ref. [5]).

B. Methods

The most time-consuming step of the experiment may be the alignment stage. The helium-neon laser should be placed into the laser mount and then turned on. Alignment of the mirrors must be done precisely and according to the standard of procedure so that the instrument can direct the laser beam through the beam splitter and mirrors properly. Instructions for the alignment stage can be found in Section 4 of Ref. [5]. It may be helpful to place a sheet of paper where the

fringe pattern is expected to appear during the initial alignment before setting up the observing camera. Once the instrument is correctly aligned, a ring of bright and dark fringes will appear on the sheet of paper like the one shown in Figure 2.

Accurate observation and measurement of the interference patterns can be done with the ThorLabs DCC1545M USB camera or the ThorLabs DET36A photodetector (See Ref. [6] for operation manual). The video feed from the camera paired with the *ImageJ* software allows for the plotting of pixel intensity against pixel location, for example. The photodetector paired with an iris blocking all but the central fringe can accurately count the number of fringes observed as the mirror is adjusted.

To measure the wavelength of the HeNe laser, the process is as follows (See Ref. [5]):

- 1) Slowly adjust the micrometer which controls the position of the adjustable mirror
- 2) Choosing a fixed point (either a pixel on the video feed or on the paper where the fringes appear), count the number of fringes which pass by that point across the range of the micrometer. Note the micrometer adjustment value each time a fringe passes this point.
- 3) Depending on the direction that the mirror is moving, circular fringes will either enter or exit at the center of the pattern.
- 4) Repeat the procedure several times to minimize errors and to obtain an average and standard deviation.
- 5) Using equation (4), calculate the laser wavelength and propagate your error to obtain an uncertainty on λ .

C. Analysis & Interpretation

In the final report, there will likely be a plot showing the number of fringes counted, N , plotted against the recorded value of Δd . This can be fitted to obtain λ . There will be uncertainty in Δd because the reading on the micrometer may be inaccurate. More importantly, it is actually not straightforward to record N since the fringe patterns are not infinitely thin. There will be some uncertainty associated with where the observer decides that another fringe has “passed”.

Slight vibrations of the instrument while data is being taken might be enough to impact the position of fringes, adding more uncertainty to the data. It is hard to account for this error quantitatively, so care must be taken not to move the interferometer while measurements are taking place.

IV. MERCURY DOUBLET

A. Theory

The Mercury lamp actually produces two spectral lines with very similar wavelengths in the visible range of the spectrum. The interferometer can be used to accurately calculate the wavelength separation between these waves, often referred to as the yellow Hg *doublet*.

The main mechanism of this experiment relies on the fact that when two waves of very similar frequency are superimposed, there are periods in which the waves are *in phase*, and their amplitudes combine coherently in what is called *constructive interference*. Of course, since there is a small

difference in wavelength, there are other regions in which the same waves are *out of phase* with each other, and their amplitudes cancel each other out in what is called *destructive interference*. This is the mechanism which creates interference fringes (bright and dark rings) which appear similar to the previous experiment.

Let's assume we have two wavelengths, λ_1 and λ_2 such that $\lambda_2 = \lambda_1 + \Delta\lambda$ and $\frac{\Delta\lambda}{\lambda_1} \ll 1$. Choosing an angle, θ , at which some dark fringe appears, and with a known difference between arm lengths d , one can write

$$2d \cos \theta = m_1 \lambda_1 = m_2 \lambda_2 \quad (5)$$

Adjusting the micrometer will bring the waves in and out of phase at θ . When the pattern returns to the high contrast state as before, the new arm length difference is d' , giving $\Delta d = d' - d$. The wavelength separation can then be calculated, given by

$$\Delta\lambda = \frac{\lambda_1 \lambda_2}{2\Delta d \cos \theta} \quad (6)$$

By looking at fringes near the center (small θ , $\cos \theta \approx 1$), equation (6) simplifies to

$$\Delta\lambda = \frac{\lambda_{avg}^2}{2\Delta d} \quad (7)$$

where λ_{avg} is the average wavelength of the doublet. (See Ref. [7]).

B. Methods

The observational instrument for this experiment should be the *ThorLabs* camera (not the photodetector) placed very close to the interferometer. The alignment process is very similar to the Helium-Neon setup, however this time the source will be a Mercury lamp instead of the previous laser. Since the lamp is less accurately approximated as a point source (and less harmful to the eye), it is helpful to look through the interferometer while adjusting the tilt controls of the mirrors so that the images of the lamp filament overlap as much as possible. Successful alignment will produce a similar bullseye pattern to before. It should appear multicolor, but mostly blue (though we are concerned with the *yellow* doublet, these are not the only spectral lines produced by the Mercury lamp).

A filter should be placed over the opening of the light source so that only the yellow doublet propagates through the interferometer. The bullseye will now be yellow.

Once a solid, high-contrast interference pattern is visible with the camera, a frame of the video feed paired with *ImageJ* can be used to capture pixel intensities to be plotted against pixel location. You should capture an image and plot it each time the micrometer is adjusted such that the pattern returns to a high contrast state. This will provide several values of Δd which can be used to calculate $\Delta\lambda$ with measured uncertainty.

As of right now, I do not know how λ_{avg} is obtained but I will find out during outline discussion.

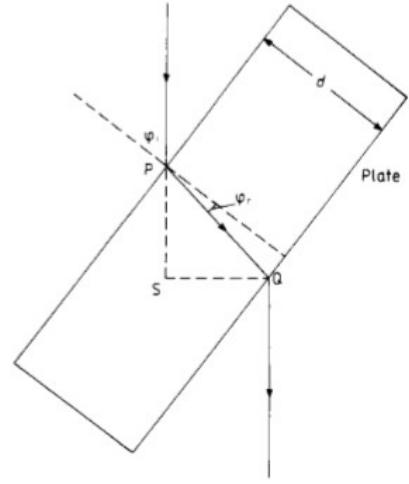


Fig. 3. Details of the light path through a plate. Knowing the thickness of the plate allows for a measurement of the refractive index. (See Ref. [8]).

C. Analysis & Interpretation

This section will probably show a few plots similar to the one shown for the "High Contrast and Low Contrast Hg Yellow Fringes" (Figure 5 in Ref. [7]). We will have to repeat the measurements several times to obtain an uncertainty on Δd , and of course there may be an uncertainty on λ_{avg} if we plan to measure this wavelength ourselves. These errors must be propagated to generate uncertainty on $\Delta\lambda$ as usual.

V. INDEX OF REFRACTION

A. Theory

Index of refraction, n , is a material property which influences how much light is bent as it passes through the boundary of a substance. It is a ratio of the apparent speed of light in vacuum to the speed of light in the material.

$$n = \frac{c}{v} \quad (8)$$

A higher refractive index thus corresponds to slower light wave propagation in the medium. At the boundary between two mediums with different values of n , the path of the light is bent. For light passing from an n_1 medium to an n_2 medium, the angle of incidence, θ_1 , and angle of refraction, θ_2 are related by Snell's law:

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_2}{n_1} \quad (9)$$

θ_1 and θ_2 are given by the angle between the path of the light and the normal direction of the interface between mediums.

Clearly, the placement of a plate such as the one in Figure 3 would produce a predictable manipulation of the path length if the plate thickness and refractive index are known. If refractive index is an unknown, then measuring the resulting path difference with the interferometer allows for a calculation of the unknown, n . This is precisely the objective for this experiment.

The refractive index of a plate placed along the path of the beam traveling between the beam splitter and the fixed mirror is given by

$$n = \frac{(2t - N\lambda)(1 - \cos \theta)}{2t(1 - \cos \theta) - N\lambda} \quad (10)$$

where λ is the laser wavelength, t is the thickness of the plate of glass, mica, or quartz, N is the number of fringes passed by, and θ is the angle rotated, in radians (see Ref. [5]).

B. Methods

Similar to the first part of this experiment (the HeNe laser), this procedure requires counting the number of fringes which pass by a particular point on the screen as the Michelson interferometer is adjusted. This time, instead of varying path length by moving the adjustable mirror with a micrometer, the rotational position of the plate in question is adjusted. This changing θ produces measurable effects on the final path length of the beam which travels through the plate, producing the expected interference patterns.

The experimental procedure is outlined as follows:

- 1) Measure the thickness of the plate of mica, quartz, or glass that is being tested.
- 2) After alignment of the interferometer so that the bullseye pattern appears on the screen, the plate should be placed in the mount.
- 3) Perform an initial alignment of the plate with the rotational adjustment handle so that it starts normal to the beam path.
- 4) Begin finely adjusting the angle of the plate. Make a note of the angle of incidence, θ , each time that a bright fringe passes the chosen fixed point on the screen, keeping track of the total number of fringes that have passed, N .
- 5) Repeat the procedure several times to obtain a statistical uncertainty to measurements.
- 6) Using the measured values of t , N , θ , and the known value of λ , perform a weighted fit to obtain the fit parameter n for the plate. Be sure to propagate uncertainties on t and θ .

C. Analysis & Interpretation

Plotting N against θ would allow for a weighted fit to produce an accurate prediction of fit parameter, n . Of course, there are several uncertainties to think about.

There may be uncertainty on the wavelength of the laser if we choose to use our value measured in part I of this experiment.

As in the first part, counting N produces some uncertainty on θ because observing when a fringe "passes" a fixed point is not straightforward since the fringes are not infinitely thin.

An exact thickness of the plate cannot be obtained, so there is uncertainty in t as well.

- 2) What is the wavelength separation of the spectral lines of the yellow Mercury doublet?
- 3) What is the refractive index of the glass, quartz, or mica plate used in the experiment?
- 4) A general discussion on the importance of this experiment will be necessary, such as the original Michelson-Morley experiment, the application of Snell's law for useful measurements, the demonstration of the wave nature of light, etc.

REFERENCES

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VI. CONCLUSIONS

Questions which will need to be answered:

- 1) What is the wavelength of the HeNe laser?